

Lost on Mars Case Study Report

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Introduction

This report summarizes my solution for the RoboTeam Twente Lost on Mars software case study. The assignment consists of two parts: designing an autonomous navigation system for a Mars rover and analyzing and improving a telemetry ring buffer implementation.

Assignment 1 – Mars Rover Design

Sensors

The proposed navigation system uses LiDAR, stereo cameras, an IMU, and wheel encoders. LiDAR provides accurate distance measurements, stereo cameras provide visual depth information, the IMU measures orientation and motion, and wheel encoders estimate travelled distance.

Sensor Fusion

Sensor data is combined using an Extended Kalman Filter (EKF). This improves localization accuracy because no individual sensor is perfect.

Mapping and Localization

SLAM (Simultaneous Localization and Mapping) is used to build a map while estimating the rover position. The map consists of an occupancy grid and elevation information to identify obstacles and rough terrain.

Navigation and Recovery

An A* path planner is used to generate safe routes. Wheel slip is detected by comparing wheel encoder data with LiDAR and camera observations. A health monitor checks sensor confidence, localization drift, and temporary sensor failures. If needed, the rover can enter a safe operating mode.

Assignment 2 – Telemetry Ring Buffer

Problem Analysis

The original ring buffer implementation used `sequenceNumber % capacity` when reading entries. This caused incorrect results after the buffer wrapped around because overwritten sequence numbers could still return data.

Solution

The solution introduces an availability check before reading data. The oldest available sequence number is calculated using `max(0, count - capacity)`. Only sequence numbers within the valid range can be accessed.

Structured Telemetry

The implementation was extended to support structured telemetry entries containing a timestamp, sensor identifier, and floating-point reading.

Testing

JUnit 5 tests were added to validate normal operation, wrap-around behavior, exception handling, telemetry entry validation, and buffer statistics. All tests passed successfully.

Conclusion

The final solution provides a complete rover navigation design and an improved telemetry ring buffer implementation. The project includes documentation, architecture diagrams, Java implementation, unit tests, and a Maven project structure.